MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

INDUCTOR, POWER, KEY CODE VPX

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic inductor module.

1.2 Classification. This specification covers one class of modules.

Class I as specified in MIL-STD-1389.

2. APPLICABLE DOCUMENTS

2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Electronic Systems Command, ATTN: ELEX 8111, Department of the Navy, Washington, DC 20360, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 0.473 failures per million operating hours when predicted in accordance with MIL-STD-1389.

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I as specified in MIL-STD-1389.

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be VPX.

3.5.2 Size. The module size shall be 1E.

3.5.3 Weight. The module shall have a maximum weight of 454 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction fin on increment A and heat dissipation fins on increments B and C required, D and E optional.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/137 -1, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance tests. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirements of MIL-M-28787.

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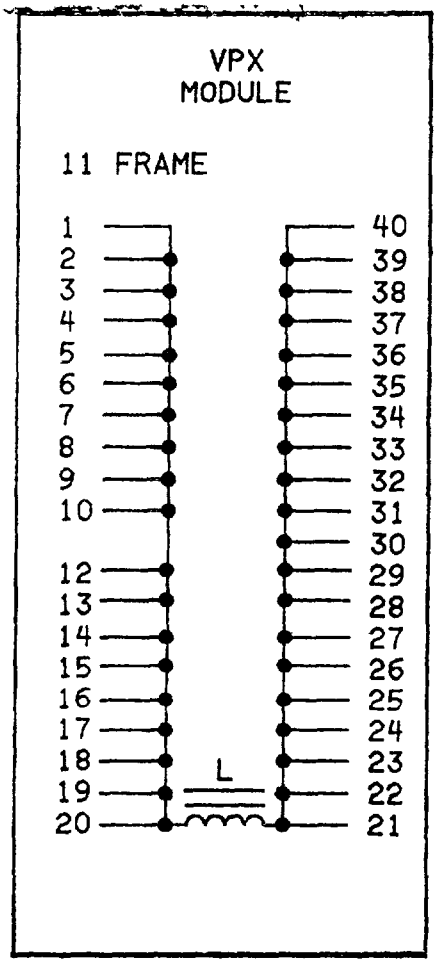


FIGURE 1. VPX module function diagram, inductor.

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TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/,2/	Acceptance requirements 25°±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
A1-A37	Continuity	II	X	0.0	0.1	0.0	0.1	0.0	0.1	0.0	0.1	Ω
B1	Inductance	III	X	300	---	300	300	---	---	300	---	μH
B2	Inductance	III	X	99	132	99	132	99	132	99	132	μH
B3	Inductance	III	X	60	---	60	---	60	---	60	---	μH
B4	Inductance	III	S	40	---	40	---	40	---	40	---	μH
C	Insulation resistance	IV	X	10	∞	10	∞	---	---	---	---	MΩ
D	Resistance check	V	S	0	0.009	0	0.009	0	0.011	0	0.011	Ω
				(0)	(0.009)	(0)	(0.009)	(0)	(0.011)	(0)	(0.011)	V
E	Frame continuity	VI	S	0	1	0	1	0	1	0	1	Ω
F1-F37	Jumper resistance	VII	S	0.00	0.02	0.00	0.02	0.00	0.02	0.00	0.02	Ω

"X" parameters shall be 100 percent tested during module acceptance tests.

"S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.

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4.4.2 Order of testing. Test conditions A through F shall be performed in any order with the following exceptions. If test condition C is a part of a sequence of tests, the order of testing shall be as follows: C where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through F in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25° ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions A through F shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature, except that test condition C shall not be tested at 0°C. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall not be operated. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column, and test conditions A, C and E shall be performed using the initial limits at 25° ±5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests A through F shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests A through F shall be performed in the order specified in 4.4.2 using the 25° ±5°C ambient end-of-life requirements of table I.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

6.3 Cross reference of classification. Military Specification MIL-M-28787/137 supersedes Department of the Navy, Naval Ordnance Systems Command Drawing No. 10001-2658682E dated 8 September 1971. Part number M28787/137-1 supersedes part numbers 2658682 and M28787/137. These parts are interchangeable.

6.4 Module description. The following electrical parameters reflect the 0° to 60°C end-of-life limits from table I.

Maximum power dissipation	4 W
Rated input voltage	2.12 Vrms, 2400 Hz
Insulation voltage rating	200 Vdc minimum
Maximum dc current	20 A
Continuous operating conditions	11 A, 6 Vp-p, 2400 Hz 11 A, 2 Vp-p, 4800 Hz

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6.5 Module function. The module functions as a single inductor (swinging choke).

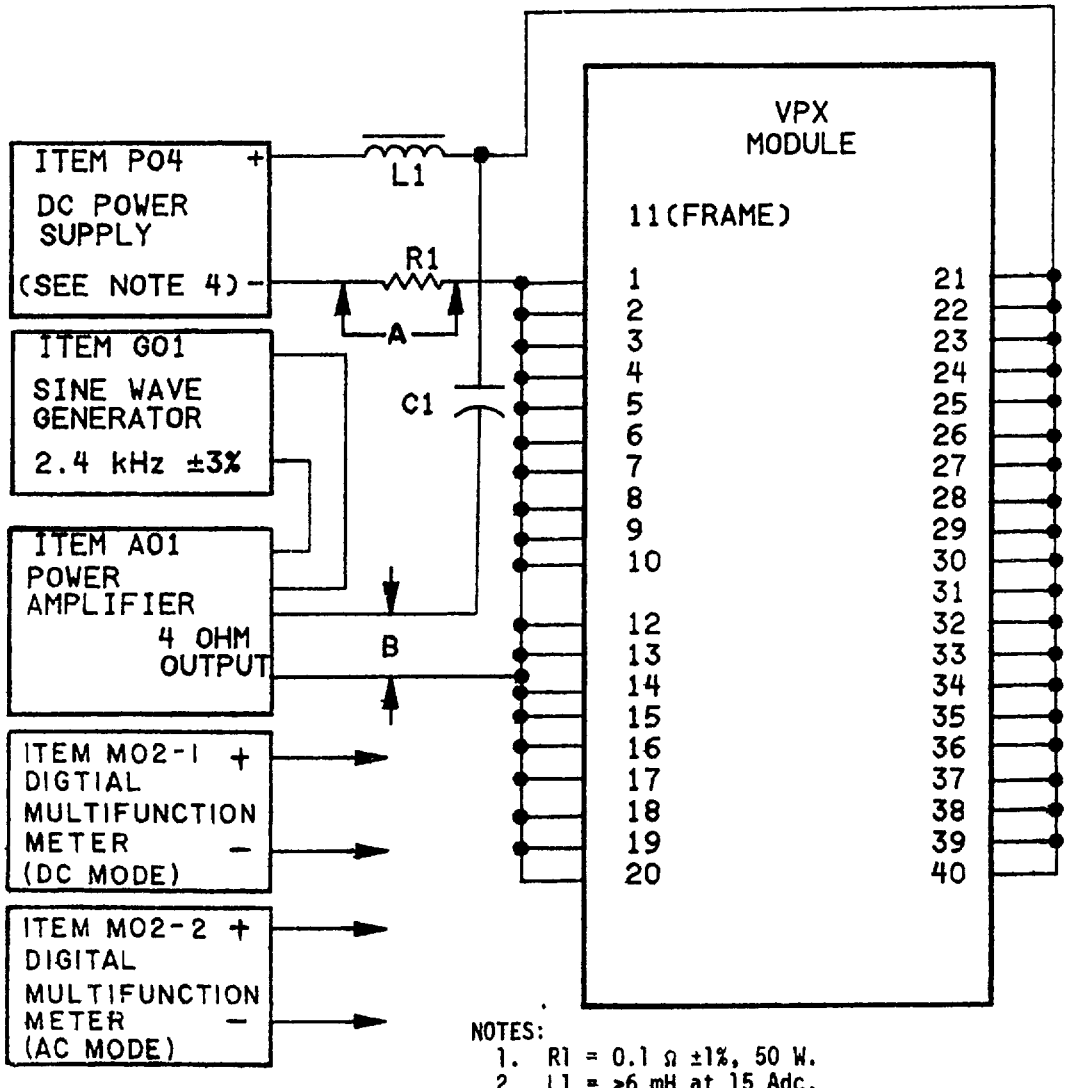
6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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NOTES:
1. R1 = 0.1 Ω ±1%, 50 W.
2. L1 = ≥6 mH at 15 Adc.
3. C1 = 10,000 µF, 10 Vdc.
4. Adjust test equipment to obtain values at:

   (a) Point A = 1.5 Vdc ±3%.
   (b) Point B = 2.12 Vrms ±3%.

FIGURE 2. Operating temperature and life test circuit.

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TABLE II. Continuity test. 1/

Test condition	Parameter	Connect Item R01		Initial limits 25°±5°C ambient		Unit
		From pin	To pin	Min	Max	
A1	Continuity	1	2	0.0	0.1	Ω
A2			3			
A3			4			
A4			5			
A5			6			
A6			7			
A7			8			
A8			9			
A9			10			
A10			12			
A11			13			
A12			14			
A13			15			
A14			16			
A15			17			
A16			18			
A17			19			
A18			20			
A19		21	22			
A20			23			
A21			24			
A22			25			
A23			26			
A24			27			

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TABLE II. Continuity test - Continued. 1/

Test condition	Parameter	Connect Item R01		Initial limits 25°±5°C ambient		Unit
		From pin	To pin	Min	Max	
A25	Continuity	21	28	0.0	0.1	Ω
A26			29			
A27			30			
A28			31			
A29			32			
A30			33			
A31			34			
A32			35			
A33			36			
A34			37			
A35			38			
A36			39			
A37			40			

1/ Each reading shall be compensated for resistance introduced by the test circuit.

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TABLE III. Inductance test.

Test condition	Parameter	Initial limits 25°±5°C ambient		Unit	Test figure
		Min	Max		
B1	Inductance	300	---	µH	3
B2		99	132		
B3		60	---		
B4		40	---		

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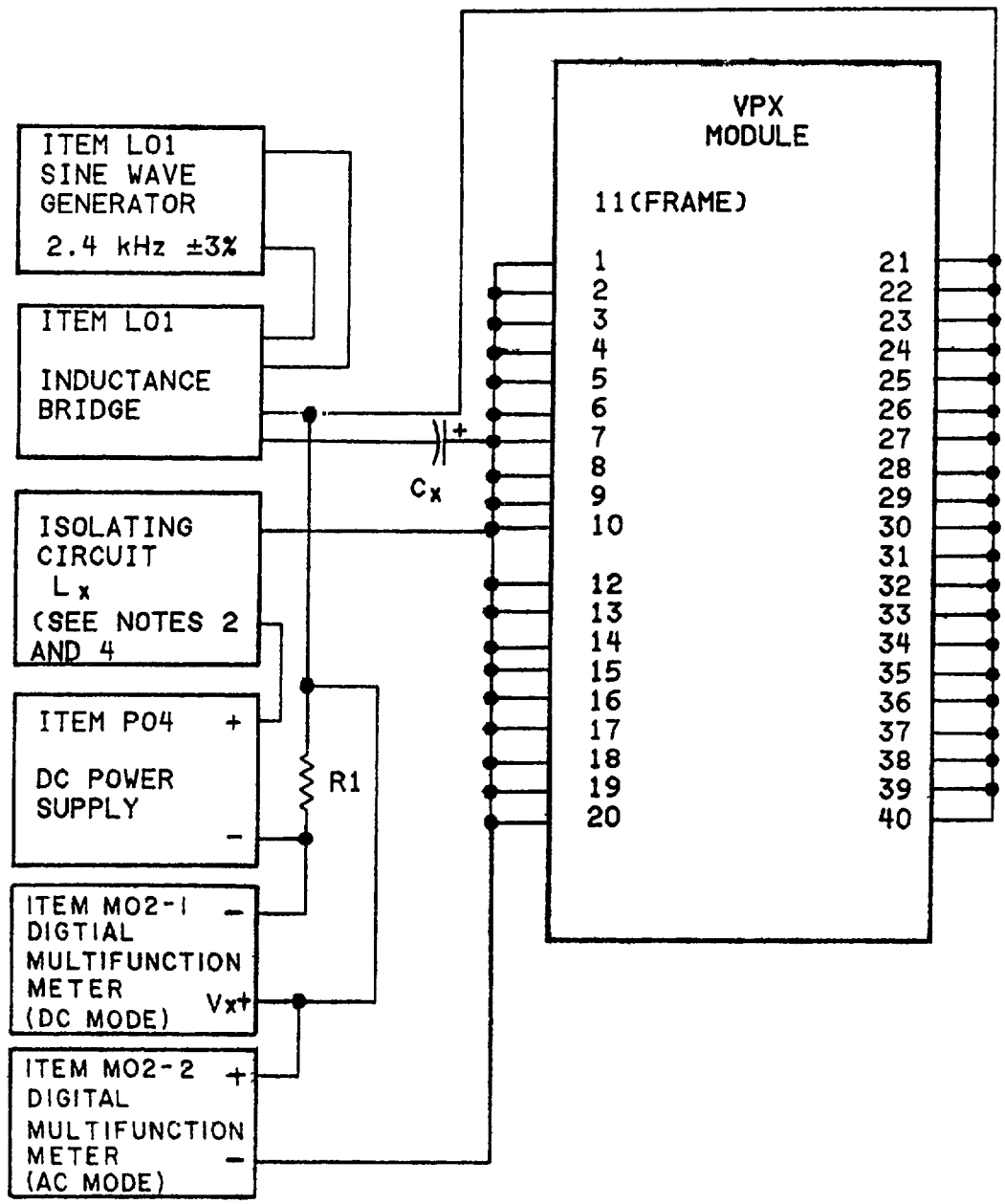


FIGURE 3. Inductance test circuit for table III.

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NOTES:
1. R1 = 0.1 Ω ±1%, 50 W
2. L_X = The isolating circuit (L_X) consists of a choke isolating circuit. Choke value is as specified for each test condition (Triad C48U and SK5650-848 or equivalent).
3. C_X = 10,000 µF 10 Vdc (use if required).
4. Use the test setup shown on figure 3 for primary magnetizing inductance test for high dc winding current. Use following values for

   (a) Test condition B1

   E_{IN} = 2 12 Vrms ±1%.
   C_X = 10,000 µF 10 V (use if required).
   L_X = 30 mH minimum, 1 Adc.
   V_X = 0 1 Vdc ±1%.

   (b) Test condition B2, same as test condition B1, except

   E_{IN} = 1.0 Vrms ±1%.
   L_X = 10 mH minimum, 10 Adc.
   V_X = 1.0 Vdc ±1%.

   (c) Test condition B3, same as test condition B1, except

   E_{IN} = 1.0 Vrms ±1%
   L_X = 6 mH minimum, 15 Adc.
   V_X = 1.5 Vdc ±1%.

   (d) Test condition B4, same as test condition B1, except

   E_{IN} = 0.7 Vrms ±1%.
   L_X = 4 mH minimum, 20 Adc.
   V_X = 2.0 Vdc ±1%.

5. Test at 20 amperes will require parallel power supplies and paralleling of the isolating inductors. Series blocking diodes may be required at the power supply outputs for protection during parallel operation (see manual for power supply used).

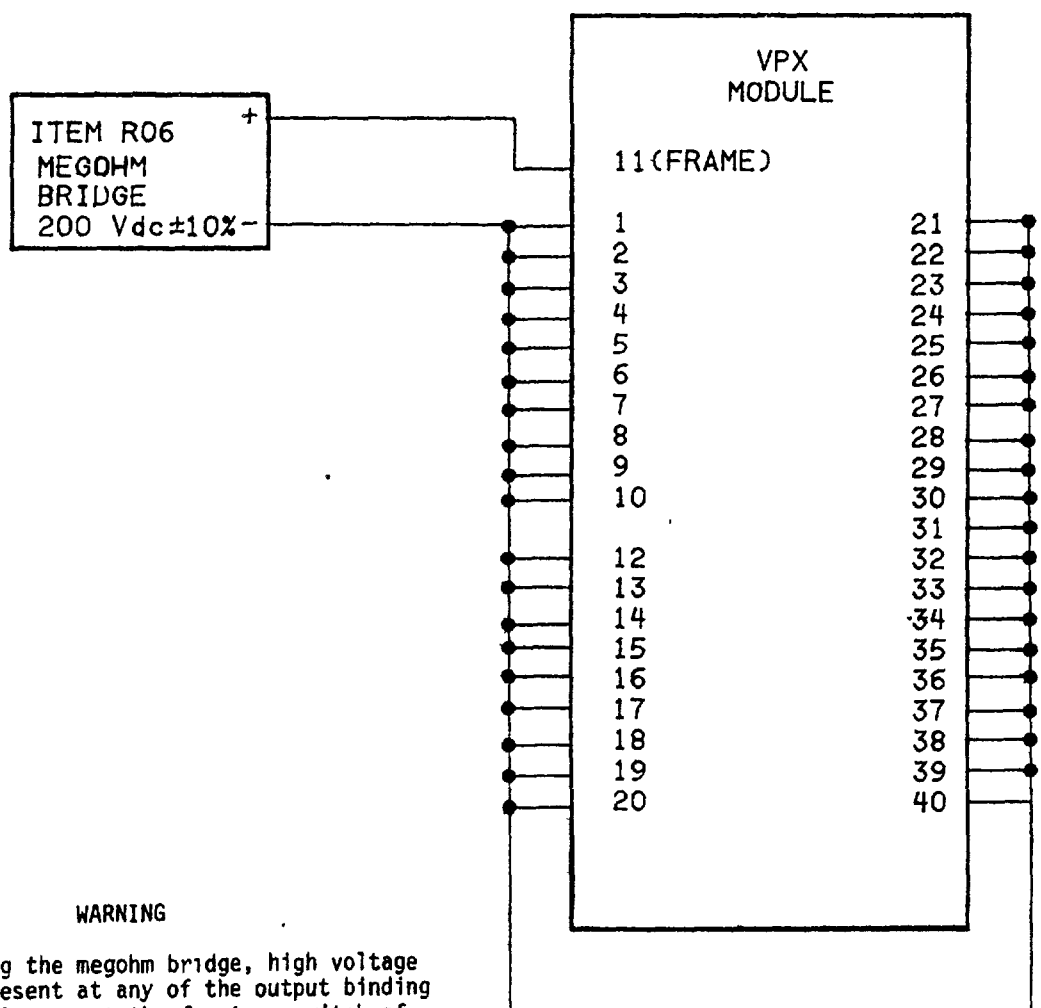
FIGURE 3. Inductance test circuit for table III - Continued.

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TABLE IV. Insulation resistance test.

Test condition	Parameter	Initial limits 25°±5°C ambient		Unit	Test figure
		Min	Max		
C	Insulation resistance	10	∞	MΩ	4

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WARNING

When using the megohm bridge, high voltage may be present at any of the output binding posts. Always set the function switch of the megohm bridge to discharge before connecting or disconnecting the test leads.

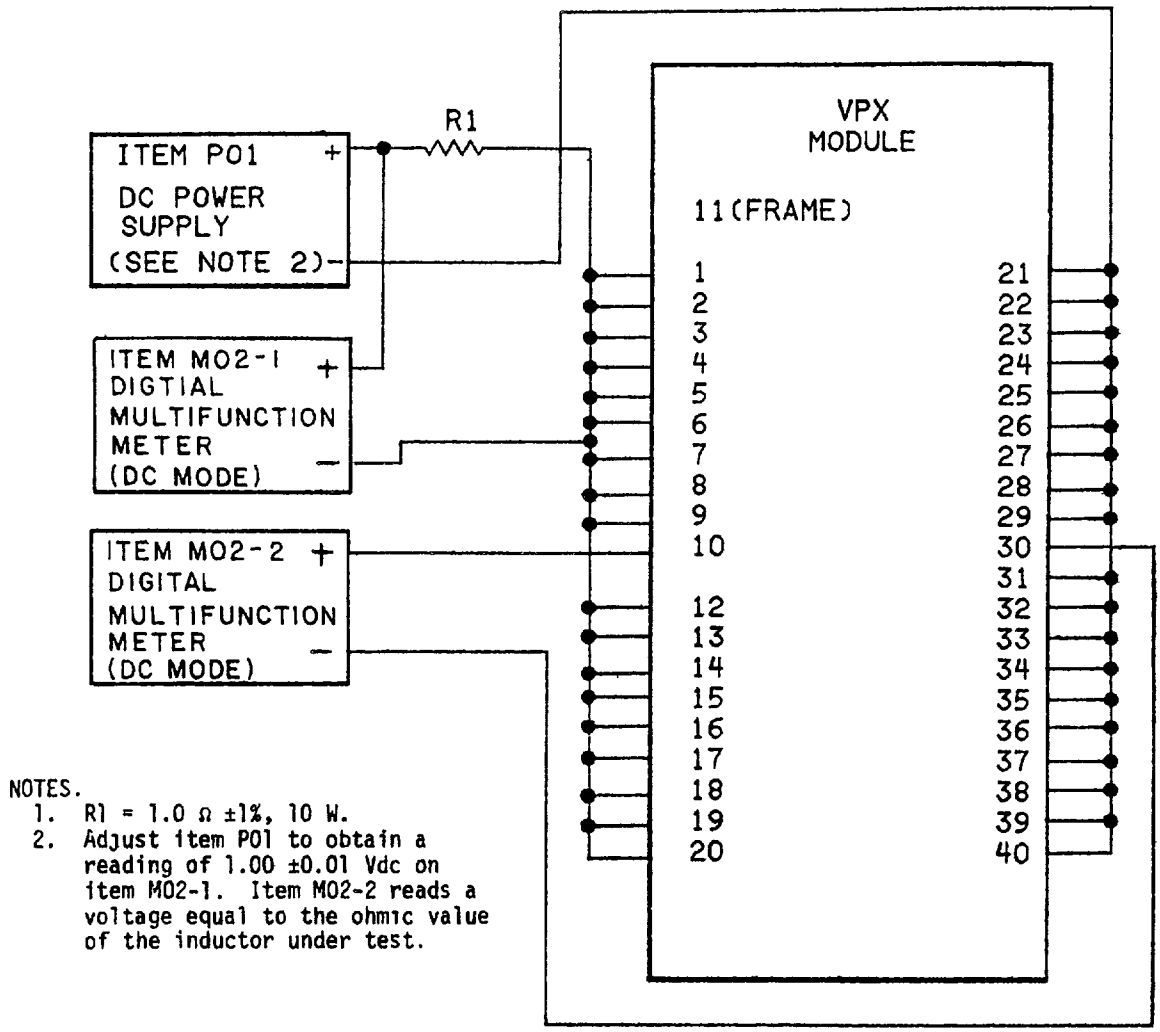
FIGURE 4. Insulation resistance test circuit for table IV.

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TABLE V. Resistance check test.

Test condition	Parameter	Initial limits 25°±5°C ambient		Unit	Test figure
		Min	Max		
D	Resistance check	0 (0)	0.009 (0.009)	V (Ω)	5

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NOTES.
1. R1 = 1.0 Ω ±1%, 10 W.
2. Adjust item P01 to obtain a reading of 1.00 ±0.01 Vdc on item M02-1. Item M02-2 reads a voltage equal to the ohmic value of the inductor under test.

FIGURE 5. Resistance check test circuit for table V.

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TABLE VI. Frame continuity test.

Test condition	Parameter	Connect Between 1/ item RO2	Initial limits 25°±5°C ambient		Unit
			Min	Max	
E	Frame continuity	Pin 11 and frame 2/	0	1	Ω

1/ The resistance between pin 11 and each fin shall be measured.

2/ Penetrate the anodized finish on the module frame with a sharply pointed probe connected to one of item RO2 leads.

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TABLE VII. Jumper resistance test. 1/

Test condition	Parameter	Connect Item R02		Initial limits 25°±5°C ambient		Unit
		From pin	To pin	Min	Max	
F1	Jumper resistance	1	2	0.00	0.02	Ω
F2			3			
F3			4			
F4			5			
F5			6			
F6			7			
F7			8			
F8			9			
F9			10			
F10			12			
F11			13			
F12			14			
F13			15			
F14			16			
F15			17			
F16			18			
F17			19			
F18			20			
F19		21	22			
F20			23			
F21			24			
F22			25			
F23			26			

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TABLE VI. Jumper resistance test - Continued. 1/

Test condition	Parameter	Connect Item R02		Initial limits 25°±5°C ambient		Unit
		From pin	To pin	Min	Max	
F24	Jumper resistance	21	27	0.00	0.02	Ω
F25			28			
F26			29			
F27			30			
F28			31			
F29			32			
F30			33			
F31			34			
F32			35			
F33			36			
F34			37			
F35			38			
F36			39			
F37			40			

1/ Each reading shall be compensated for resistance introduced by the test circuit.

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